

Incident report analysis

Summary	The organization experienced a Denial-of-Service (DoS) attack caused by a flood of incoming ICMP echo request packets (pings). A misconfigured firewall allowed malicious traffic to overwhelm the internal network, making network resources unavailable for two hours. The incident response team blocked incoming ICMP packets, shut down non-critical network services, and restored critical services. After investigating the incident, the security team implemented additional firewall rules, IP source address validation, network monitoring software, and an IDS/IPS to reduce the risk of future attacks.
Identify	• **Attack type:** Denial-of-Service (DoS) attack using an ICMP flood. • **Affected assets:** The internal network, network services, and resources that legitimate users needed to access. • **Vulnerability:** A misconfigured firewall allowed excessive incoming ICMP traffic. • **Business impact:** Employees could not access internal network resources for two hours, disrupting the organization's web design, graphic design, and social media marketing services. • **Investigation:** Review firewall configurations, network logs, ICMP traffic patterns, and the source addresses associated with the attack.
Protect	• Configure firewall rules to limit the rate of incoming ICMP packets. • Validate source IP addresses and block spoofed or untrusted traffic. • Review firewall configurations regularly to identify security weaknesses. • Keep network devices and security software updated. • Use an IDS/IPS to identify and filter suspicious traffic. • Train IT staff to recognize network threats and follow security

	procedures.
Detect	• Continuously monitor network traffic for unusual increases in ICMP packets. • Configure alerts for abnormal traffic volumes and potential DoS attacks. • Use network monitoring software to identify suspicious traffic patterns. • Review firewall and network device logs. • Use an IDS to detect suspicious activity and an IPS to block selected malicious traffic. • Consider using a SIEM to centralize logs and correlate security events.
Respond	• Activate the incident response plan and notify the security team. • Apply firewall rules to limit or block malicious ICMP traffic. • Isolate affected network resources when necessary while preserving critical services. • Review firewall logs, network traffic records, and source IP addresses to investigate the incident. • Communicate the incident and its impact to relevant IT staff and management. • Document the actions taken and update incident response procedures to prevent similar incidents.
Recover	• Verify that malicious traffic is under control before restoring normal operations. • Restore critical network services and gradually bring non-critical services back online. • Test network connectivity and confirm that authorized users can access required resources. • Monitor the network closely after recovery to identify any recurring

	suspicious activity. ● Inform employees and management when services are restored. ● Review the recovery process and update security controls and procedures based on lessons learned.

Reflections/Notes: This incident demonstrates the importance of correctly configured firewalls, continuous network monitoring, and effective incident response procedures. The NIST Cybersecurity Framework helps organizations identify vulnerabilities, protect assets, detect threats, respond to incidents, and recover normal operations. Regular security reviews and improvements can reduce the impact of future DoS attacks.